



DEPARTMENT OF COMPUTER TECHNOLOGY

Template and Questions on Group Technical Reaction – S15

Term 3 AY 2025-2026

Group Number 5 Score/Rubrics based Group Technical reaction.	Group Members & Roles: 1. __SANCHEZ SAMANTHA BREANNE AVILA _____ (Lead Analyst) 2. __PAJE SHADRACH PAUL CHICHIOCO_____ (ITSM/ITIL Specialist) 3. _Dana Pelagio _____ (COBIT Governance Auditor) 4. __AGUNANNE HENRY CHINEMEREM_____ (DevOps Integration Architect)
Case study reference : <i>Manufacturing Change Management - A Case Study-Based Analysis Of Change Processes In Industrial Practice (Rammo, Mujcinovic, & Zaeh, 2024)</i>	Date of Submission: 10/09/2026

I . Executive Summary

(Provide a concise overview of the reaction paper, synthesizing how the Rammo et al. (2024) case study illustrates the necessity of adapting Manufacturing Change Management (MCM) from an isolated, localized operational fix into an integrated, enterprise-wide Service Management strategy. Detail how the varying levels of maturity across Company A, B, and C reveal critical operational gaps when relying on fragmented tools like Excel versus automated workflow engines. Explain how modern Information Technology Service Management (ITSM) principles can reconcile the tension between rapid operational agility and robust governance control. Limit to 250 words.)



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Rammo et al. (2024) illustrated that effective Manufacturing Change Management must evolve from localized operational fixes into an integrated, enterprise-wide strategy to stay competitive in volatile environments. The study reveals that relying on tools like Excel creates significant operational gaps. Company C, with low maturity, suffers from department-specific, inaccessible data due to a lack of formal systems. Company A faces a critical gap where perceived complexity leads to underutilization and a retreat to Excel and email, resulting in unreliable data where 25% of changes are inaccurately tracked. In contrast, Company B's high maturity is defined by its use of an automated workflow engine. This tool provides a robust database with meticulous timestamps, allowing for advanced optimization methods like process mining. These findings suggest that modern ITSM principles can reconcile the tension between agility and governance. By replacing fragmented spreadsheets with standardized, automated workflows, organizations achieve the governance control necessary for regulatory compliance while reducing the manual time expenditure that often drives users toward shadow IT. Ultimately, transitioning MCM into a service-oriented framework ensures that individual processes are adaptable yet systematically controlled, enabling efficient handling of increasing change variety.

II. Learning Objectives

- **Apply** structured ITIL Change Enablement workflows, automated configuration tracking, and parameterized routing rules to optimize volatile enterprise change environments.
- **Apply** governance framework controls (such as COBIT BAI06) and process mapping notations (BPMN 2.0) to model, track, and audit cross-functional organizational lifecycles.
- **Analyze** the operational, cultural, and systemic friction points between highly complex process documentation and the real-world compliance rates of frontline users

III Phase A: Application-Level Assessment Matrix (Questions 1–10)

Q1. Application of ITIL Change Enablement Roles to Decentralized Operational Models

Company C currently lacks a dedicated change management department, relying instead on individual department heads to act as ad-hoc requestors and managers. Apply ITIL Change Enablement practices to design a standardized, centralized



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role-allocation framework for Company C. Demonstrate how establishing clear boundaries between the Change Initiator, Change Reviewer, and Emergency Change Authority prevents conflicts of interest and ensures objective risk assessments.

The organization must implement a centralized role-allocation framework that designates specific duties to a Change Initiator, a Change Reviewer, and an Emergency Change Authority. Currently, Company C relies on department heads to act as ad-hoc managers, leading to department-specific data that is often inaccessible. By assigning project managers as Initiators and establishing a neutral Reviewer role, the company ensures that changes are evaluated against enterprise-wide standards rather than localized needs. This separation of roles creates a vital check and balance system that prevents conflicts of interest, as the party requesting the change no longer holds the sole authority to approve or document it, ensuring more objective risk and cost assessments. Ultimately, this framework replaces undocumented email communication with a centralized database, providing the governance control required for regulatory compliance while the Emergency Change Authority preserves the operational agility needed to manage a high variety of manufacturing changes.

Q2. Designing an ITIL-Compliant Emergency Change Workflow for Regulatory Shifts

In the medical technology sector (Company B), regulatory compliance is a key driver of operational change. Apply ITIL Service Management concepts to design an accelerated **Emergency Change Workflow** that handles a sudden regulatory compliance failure without bypassing vital validation testing. Your design must trace the request path from logging through immediate technical validation to post-implementation review.

An ITIL-compliant Emergency Change Workflow for regulatory shifts begins with the Requestor logging an Emergency Change Request directly into the end-to-end workflow tool, which provides immediate, meticulous timestamps for auditability. The path then moves to immediate technical validation, where the Change Manager and Reviewer perform an accelerated impact assessment. Unlike standard changes that average 185 days, this emergency path front-loads validation testing by utilizing existing creativity techniques and cost-utility analysis to ensure the solution meets regulatory standards without bypassing safety protocols. Following implementation, which is managed using the tool's automated task distribution, the workflow concludes with a



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Post-Implementation Review. During this final stage, the organization leverages its robust database and process mining to analyze the emergency execution for inefficiencies, ensuring the results are verified in internal discussion rounds to maintain long-term factory stability and compliance.

Q3. Constructing an API-Driven Workflow to Prevent "Identity Drift" in Completed Changes

Company A suffers from a data integrity gap where over 25% of changes remain marked as "in process" within the ERP system despite being physically completed months prior. Apply ITSM workflow engineering principles to design an automated API validation bridge. This bridge must connect the physical factory floor confirmation point back to the central ERP system, automatically closing the change file once specific closing parameters are satisfied.

To address the identity drift at Company A, where over 25% of changes remain inaccurately marked as "in process" due to ERP underutilization and a reliance on manual Excel updates, an automated API validation bridge should be implemented to link physical factory floor completion directly to the central system. Drawing on ITSM workflow engineering, this bridge initiates an automated closing sequence once a physical confirmation point is triggered. The API then validates that specific closing parameters are satisfied, including the verification of successful validation testing, the upload of final change release documentation, and the completion of all 78 defined process steps. By automating this, the system adopts the high maturity end-to-end workflow characteristics seen in Company B, which utilizes meticulous timestamps to maintain a robust and reliable database. This effectively eliminates the perceived complexity and time expenditure that currently drives users toward unreliable shadow IT, ensuring the ERP system accurately reflects the physical state of the factory in real-time.

Q4. Operationalizing BPMN 2.0 Standards to Standardize Cross-Functional Information Flows

The case study highlights significant heterogeneity in how companies model their change steps, switching between detailed BPMN 2.0 layouts, Event-Driven Process Chains (EPC), or no documentation at all. Apply BPMN 2.0 notation standards to build a uniform blueprint for a cross-functional change review process. The blueprint must clearly map swimlanes for Procurement, Engineering, and Quality Management across three core phases: initial request logging, cross-department impact analysis, and final sign-off.



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To operationalize a uniform BPMN 2.0 blueprint for cross-functional Manufacturing Change Management (MCM), organizations must integrate swimlanes for Engineering, Procurement, and Quality Management across three core phases to eliminate the "department-specific" data silos identified in the sources. In the initial request logging phase, the Engineering swimlane serves as the "Requestor," capturing the change cause, such as quality improvements, supplier demands, or regulatory shifts, into a formal digital document. This transitions into a cross-department impact analysis, where a parallel gateway triggers simultaneous evaluations: Engineering performs technical risk and stakeholder analyses, Procurement conducts cost-utility evaluations, and Quality Management ensures the change aligns with regulatory requirements. The final sign-off phase involves mandatory validation testing and the issuance of a "Change Release" by a neutral Reviewer or Change Manager, ensuring that 100% of changes are accurately tracked with meticulous timestamps. By replacing fragmented, email-based communication with this systematic BPMN 2.0 standard, companies can achieve the high-maturity "end-to-end workflow" necessary to maintain a robust database for process mining and continuous improvement.

Q5. Implementing COBIT BAI06.01 (Evaluate, Prioritize and Authorize Change Requests) Controls

Apply the core control requirements of COBIT BAI06.01 to Company A's "change rounds" system. Demonstrate how an IT or operational auditor would replace subjective expert estimates with a standardized, risk-weighted evaluation matrix. This matrix must explicitly calculate financial, operational, and scheduling risks prior to grant-authorizing a change request.

To implement COBIT BAI06.01 controls at Company A, an auditor would transition the organization from its informal "change rounds" governed by subjective expert estimates into a standardized, risk-weighted evaluation matrix integrated directly into the ERP system. This matrix would replace the current unreliable data, where 25% of changes remain incorrectly marked as "in process" by objectively calculating financial, operational, and scheduling risks prior to authorizing any of the approximately 159 annual changes. Financial risk would be quantified by weighting projected expenditures against specific "supplier" or "customer demands" using ERP data, while operational risk would evaluate the impact on the 78 defined process steps and existing factory elements to identify alterations requiring rigorous validation testing. Finally, scheduling risk would use Company A's 106-day average lead time as a benchmark to ensure new requests do not exceed capacity or drive users toward shadow IT due to perceived complexity. This data-driven framework ensures that authorization is a formal gate,



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maintaining long-term factory stability rather than a byproduct of informal departmental discussions.

Q6. Creating a RACI Matrix to Manage Complex Multi-Role Change Environments

Company B's change management process involves eight distinct operational roles. Apply the structural design of a RACI Matrix to map out responsibilities across four core process milestones: initial technical verification, cost-benefit analysis, deployment testing, and configuration management database (CMDB) baseline updates. Ensure that accountability remains constrained to a single role per milestone to prevent ownership conflicts.

Process	Responsible	Accountable	Consulted	Informed
Technical Verification	Technical Engineer	Change Manager	QA Team	Project Team
Cost-Benefit Analysis	Finance Analyst	Change Manager	Engineering	Stakeholders
Deployment Testing	QA Lead	QA Manager	Operations	Service Desk
CMDB Update	Configuration Manager	Configuration Manager	Change Manager	All Teams

Assigning one accountable role per milestone prevents ownership conflicts while ensuring collaboration across departments.

Q7. Utilizing Change Lead Time Telemetry for Predictive Service Capacity Planning

Company B experiences an average change lead time of 185 days, while Company A averages 106 days. Apply Service Level Management (SLM) trend analysis techniques to establish an operational capacity dashboard. The design must parse change logging histories and status timestamps to flag changes that exceed normal duration targets, alerting management before delivery timelines are impacted.



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A Service Level Management dashboard should track average lead times, overdue approvals, implementation duration, and pending reviews. Automated alerts trigger when a change exceeds predefined thresholds (e.g., 120 days). Trend analysis identifies recurring bottlenecks by department, change type, or approver. Managers can then intervene before delays affect production schedules. Historical reporting also supports continual service improvement by identifying process inefficiencies and measuring performance against service targets.

Q8. Designing an Automated Configuration Item (CI) Verification Routine for Edge Environments

Company C manages its change documentation using fragmented local spreadsheets stored across disjointed network folders. Apply ITIL Service Configuration Management concepts to design a technical execution plan for an automated configuration verification engine. This engine must periodically crawl network directories, parse localized data attributes, and automatically sync them into a single, reliable Configuration Management Database (CMDB).

Company C should deploy an automated discovery tool that periodically scans network folders, extracts spreadsheet data, validates Configuration Item attributes, and synchronizes verified information into a centralized CMDB. Duplicate entries, missing fields, and inconsistent records should generate alerts for review. Scheduled synchronization improves data consistency while reducing manual updates. Centralized configuration information strengthens impact analysis, supports audits, and provides reliable visibility across departments.

Q9. Operationalizing COBIT DSS05.02 (Manage malware and security vulnerabilities) in Cloud Workflows

Company B uses an end-to-end cloud workflow tool to distribute tasks and communicate change updates across its operations. Apply the specific data security controls of COBIT DSS05.02 to this platform. Design a secure configuration path that enforces strict data isolation, end-to-end payload encryption, and conditional access policies to protect intellectual property and proprietary medical designs from external exploits.

Company B should enforce role-based access control, multi-factor authentication, end-to-end encryption, and network segmentation for its cloud workflow platform. Sensitive engineering documents should remain isolated through least-privilege permissions, while continuous vulnerability scanning and security monitoring detect threats early. Audit logs should record all user activities to support compliance



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investigations. These DSS05.02 controls protect intellectual property and reduce cybersecurity risks without disrupting collaboration.

Q10. Formulating Service Desk Triage Scripts for High-Volume Supplier Requests

At Company A, supplier-driven demands represent a major source of operational change requests. Apply ITIL Request Management principles to write an automated triage script for the IT Service Desk portal. This script must analyze incoming supplier requests, extract key cost and material impact data, and automatically assign a priority tier before routing the request to the correct engineering team.

The automated portal should capture supplier name, affected materials, estimated cost, production impact, urgency, and supporting documents. Rules classify requests into Low, Medium, High, or Critical priority based on business impact and urgency. Supplier requests affecting production or compliance receive immediate routing to Engineering, while lower-risk requests enter the standard review queue. Automated categorization reduces manual workload, accelerates response times, and ensures consistent prioritization across supplier-driven changes.

IV Phase B: Analysis-Level Assessment Matrix (Questions 11–20)

Q11. Deconstructing Process Friction: High Complexity vs. Low User Adoption

Analyze the systemic friction that occurs when an enterprise deploys an overly complex ERP change tracking workflow that leads users to bypass it in favor of unauthorized Excel tracking sheets. Differentiate between the nominal governance benefits of a highly detailed 78-step process design and the actual security and data visibility risks introduced by the resulting "shadow IT" workarounds.

Company A's detailed 78-step ERP workflow provides strong governance, traceability, and compliance. However, its complexity encouraged employees to use Excel spreadsheets instead, creating "shadow IT." This bypass weakened data integrity, reduced visibility, and increased security risks because changes were no longer centrally tracked. A balance between governance and usability is essential. Simplifying workflows and automating repetitive tasks would improve user adoption while maintaining compliance and auditability.



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Q12. Structural Breakdown of Data Fragmentations in Decentralized Change Operations

Analyze the structural impact on organizational data integrity when a company relies on isolated personal emails and local network folders to coordinate cross-departmental changes. Evaluate how a lack of centralized data management limits data accessibility, creates information silos, and hinders root-cause analysis during post-mortem audits.

Using personal emails and local folders creates disconnected information silos where change records become inconsistent and difficult to retrieve. This limits collaboration between departments, increases duplicate work, and complicates root-cause analysis during audits or incidents. Without centralized data management, organizations struggle to identify historical trends and accurately measure performance. A unified workflow platform and centralized repository improve accessibility, transparency, and decision-making across the entire organization.

Q13. Dissecting the Tension Between Rapid Operational Adjustments and Regulatory Auditing

Analyze the compliance friction that emerges within high-maturity sectors (such as Company B's medical technology operations) when trying to balance immediate, on-the-floor changes with rigid regulatory documentation mandates. Explain how manual verification workflows can delay critical updates, and analyze how automated, software-driven tracing helps resolve this operational bottleneck.

Medical technology companies must implement operational changes quickly while maintaining strict regulatory documentation. Manual approvals and verification often delay implementation despite urgent needs. Automated workflow systems address this by recording approvals, timestamps, and validation results in real time. This preserves compliance without sacrificing responsiveness. Digital traceability also simplifies audits, reduces documentation errors, and ensures every regulatory requirement is consistently documented throughout the change lifecycle.

Q14. Evaluating Governance Framework Strengths: ITIL Change Enablement vs. COBIT BAI06



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Analyze how the case study's findings map to international IT governance standards. Contrast the day-to-day operational focus of ITIL Change Enablement practices (handling individual requests, tracking tasks) with the broader governance controls of COBIT BAI06 (Managed IT Changes). Determine which framework provides better structural protections when an enterprise faces rapid, unpredictable changes in a volatile business environment.

ITIL Change Enablement focuses on managing individual changes through planning, approval, implementation, and review to minimize service disruption. COBIT BAI06 emphasizes governance by defining organizational controls, accountability, and risk management for all IT changes. In volatile environments, both frameworks complement each other. ITIL improves operational efficiency, while COBIT ensures strategic oversight and compliance. Together, they provide stronger protection than either framework alone.

Q15. Analyzing "Identity Drift" in Shared Configurations and Stale Process States

Analyze the operational risks and financial implications of a system where over a quarter of documented changes remain stuck in an "in process" status long after physical deployment is complete. Explain how these unresolved states distort capacity metrics, undermine the reliability of the corporate CMDB, and expose the organization to compliance failures during third-party audits.

Leaving completed changes marked as "in process" produces inaccurate performance metrics and unreliable CMDB records. Management may overestimate active workloads, delaying resource allocation and decision-making. Outdated records also complicate compliance audits because documented system states no longer match actual operations. Automating status updates and enforcing workflow completion improve reporting accuracy, strengthen governance, and reduce financial risks caused by poor operational visibility.

Q16. Dissecting the Relationship Between Process Mining Telemetry and Process Optimization

Company B tracks all process steps with detailed timestamps and clear role assignments within its workflow engine. Analyze how an organization can use process mining techniques on this event log data to uncover hidden bottlenecks. Detail the specific ways that comparing planned timelines with actual execution histories can reveal hidden inefficiencies in approval cycles.



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Company B's detailed workflow logs allow process mining to compare planned workflows with actual execution. Timestamp analysis identifies approval bottlenecks, repeated rework, and departments causing delays. Organizations can then redesign inefficient activities, automate repetitive approvals, or redistribute workloads. Process mining supports continual improvement by replacing assumptions with measurable evidence, enabling data-driven optimization of change management performance.

Q17. Evaluating the Suitability of Ad-Hoc Cost Assessments Against Formal Governance Models

At Company A, changes are evaluated during periodic "change rounds" using quick, expert-estimated cost and risk profiles. Analyze the structural vulnerabilities of relying on these subjective assessments compared to formalized, data-driven financial management models. Explain how subjective estimates can lead to unexpected budget overruns and poor resource allocation across long-term initiatives.

Expert judgment is valuable but may vary between reviewers and overlook hidden costs. Subjective estimates increase the risk of inaccurate budgets, inconsistent prioritization, and inefficient resource allocation. Formal governance models use standardized financial metrics, historical project data, and quantitative risk assessments to improve decision-making. Combining expert knowledge with objective evaluation creates more reliable and transparent change approval processes.

Q18. Strategic Evaluation of Process Standardization Across Diverse Global Portfolios

The case study notes that change processes vary significantly based on industry demands, regulatory rules, and company size. Analyze the core challenges an enterprise faces when trying to roll out a single, rigid corporate change framework across highly diverse business units. Evaluate the balance between enforcing global process standards and allowing local units the flexibility to adapt to their specific market realities.

A single global change process promotes consistency, easier auditing, and standardized reporting. However, different industries, regulations, and operational requirements make rigid implementation impractical. Organizations should establish a common governance framework while allowing local departments to customize workflows within approved boundaries. This approach maintains enterprise-wide control without reducing operational flexibility or responsiveness to local business conditions.



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Q19. Deconstructing the Vulnerabilities of Custom-Built Tools in Consolidated Environments

Company B uses a custom, in-house application specifically for rough cost estimations, alongside an enterprise workflow engine. Analyze the technical and systemic risks of maintaining these isolated, single-purpose calculation tools within an overall change infrastructure. Focus on issues related to data synchronization errors, long-term support costs, and the lack of real-time visibility for executive management.

Standalone cost estimation tools may not synchronize automatically with enterprise workflow systems, causing inconsistent information across platforms. These tools also require ongoing maintenance, increasing long-term support costs. Limited integration reduces executive visibility because financial data cannot be monitored in real time. Integrating custom applications through APIs or replacing them with enterprise modules improves consistency, reporting accuracy, and operational efficiency.

Q20. Dissecting Cultural Resistance to Automated Workflow System Migrations

Analyze the cultural and organizational factors that cause teams to resist moving from familiar, informal tracking methods (like email and spreadsheets) to structured, automated enterprise workflow tools. Dissect the underlying drivers of this resistance, and analyze how target change management strategies can help teams view structured platforms as tools that save time rather than as restrictive management overhead.

Employees often resist automated systems because they perceive them as complicated, restrictive, or slower than familiar spreadsheets and email. Poor usability and insufficient training reinforce this resistance. Effective organizational change management addresses these concerns through user training, simplified interfaces, executive support, and clear communication of benefits. Demonstrating how automation reduces repetitive work and improves collaboration increases acceptance and long-term adoption.

V Technical Reaction, Evaluation & Findings

(The group must document their core technical analysis of the case study here. Critically evaluate the findings across the three companies, detailing whether you agree or disagree with the teams' choices to supplement or completely substitute enterprise systems with spreadsheet workarounds to bypass complex processes.)



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The case study demonstrates that successful Manufacturing Change Management depends on balancing governance with practical usability. While Companies A and B implemented structured processes, Company A's overly complex ERP workflow encouraged spreadsheet workarounds that weakened data integrity and governance. Company B's integrated workflow engine provided stronger visibility, accountability, and auditability. Company C showed that decentralized management limits consistency and organizational learning. We disagree with replacing enterprise systems with spreadsheets because they introduce shadow IT, fragmented data, and compliance risks. Instead, organizations should simplify enterprise workflows and automate repetitive tasks to encourage user adoption while maintaining governance.

VI. Governance & Process Recommendations

(Formulate structural engineering and governance recommendations based on your analysis. Recommendations must align with COBIT BAI06 change lifecycles and ITIL Continual Service Improvement models.)

[Example Framework Mapping Structure]

Recommendation -> Target ITIL Practice -> Target COBIT Control -> Expected Outcome

Recommendation	Target Practice ITIL	Target COBIT Control	Expected Outcome
Simplify approval workflows	Change Enablement	BAI06	Higher user adoption and fewer manual workarounds
Implement centralized workflow software	Service Configuration Management	BAI06	Improved traceability and audit readiness
Automate ERP and CMDB synchronization	Configuration Management	BAI09	Accurate, real-time configuration data



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Develop KPI dashboards for lead times	Continual Improvement	MEA01	Early detection of bottlenecks and improved performance
Conduct regular process reviews	Continual Improvement	APO11	Continuous optimization and stronger governance

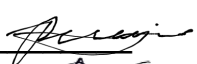
VII. Conclusion & Lessons Learned

(Summarize the group's insights regarding how modern service providers can transform traditional change processes into strategic business assets, maintain strong regulatory compliance during volatile market shifts, and leverage automated workflow tools to protect data integrity and operational safety.)

The study highlights that effective Manufacturing Change Management requires both structured governance and user-friendly processes. Companies with integrated workflows achieved better visibility, accountability, and regulatory compliance than those relying on manual coordination. However, excessive process complexity can discourage adoption and lead to shadow IT practices. Applying ITIL Change Enablement alongside COBIT governance enables organizations to standardize change processes, improve data integrity, and support continual improvement. Ultimately, digital workflows, automation, and balanced governance transform change management from a reactive operational activity into a strategic capability that strengthens organizational resilience and long-term competitiveness.

VIII . Verification & Sign-off Block

We, the undersigned members of Group 5, hereby certify that this technical reaction paper represents a collaborative team analysis. Every member contributed to the systemic application and analysis of the target frameworks against the provided case study context.

- **Lead Change Analyst Signature:**  Date: 09/07/2026
- **Service Specialist Signature:**  Date: 09/07/2026
- **Governance Auditor Signature:**  Date: 09/07/2026
- **Workflow Architect Signature:**  Date: 09/07/2026